QvsView.qs

Read-only Qlik load script viewer with syntax highlighting — add it to any Qlik Sense sheet to display a field containing app load scripts in a formatted, color-coded code viewer.

Primarily intended for use with **Qlik Sense Enterprise on Windows** (client-managed), but also compatible with **Qlik Sense Cloud**.

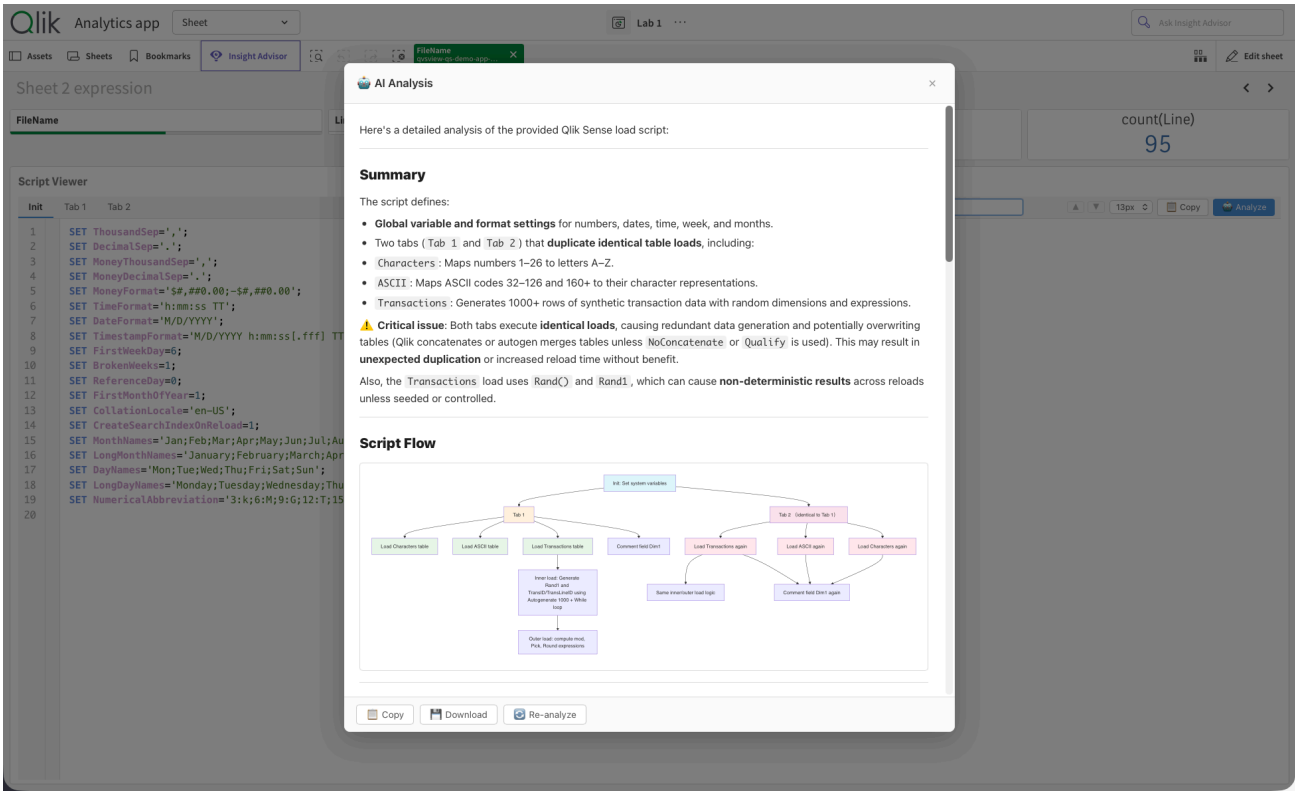
❤️ Support the project

If you find this project helpful and use it in your Qlik Sense environment, please consider supporting it financially! Your sponsorship helps ensure the project's long-term sustainability and allows me to continue maintaining it, fixing bugs, and adding new features.

👉 [Sponsor the project on GitHub](#) — click the "Sponsor" button at the repository page to become a sponsor.

- ⭐ **Star the repository** on GitHub — it helps others discover the project
- 🍴 **Fork and contribute** — pull requests are welcome!
- 💬 **Share your feedback** — let me know how you're using it
- 🐛 **Report issues** — help improve stability and functionality

This project is maintained by [Göran Sander](#) and supported by [Ptarmigan Labs](#).



Features

- **Syntax highlighting** — color-coded keywords, functions, strings, comments, variables, and field references sourced directly from Qlik's own BNF grammar (~400 keywords and functions across 25 categories, matching the native script editor's color scheme)
- **Section tabs** — automatically detects script sections and renders a clickable tab strip for instant navigation between script sections

- **Search** (Ctrl/Cmd+F) — real-time search with highlighted matches; automatically expands any folded regions that contain a match
- **Code folding** — collapse and expand sections to focus on specific parts of the script; fold state is preserved while navigating
- **Copy to clipboard** — one-click copy of the visible section or the full concatenated script
- **Deprecated function detection** — deprecated Qlik functions are rendered with strikethrough styling, sourced from the BNF definition
- **Runtime BNF** — optionally fetches the authoritative keyword list live from the Qlik Engine API for maximum accuracy
- **AI script analysis** — send the script to Ollama, OpenAI, or Anthropic for AI-powered review; four prompt templates (General, Security, Performance, Documentation); output rendered as Markdown with Mermaid diagrams; results cached in `sessionStorage` for 30 minutes
- **Large script support** — paginated `getHyperCubeData` fetching handles arbitrarily large scripts without truncation
- **Multi-app detection** — warns when dimension data comes from multiple apps (the viewer is designed to show a single app's script)

Getting Started

Prerequisites

- **Qlik Sense Enterprise on Windows** (client-managed) — May 2025 or later (may work on older versions but is not tested) or **Qlik Sense Cloud**
- Two fields in your data model:
  - One field containing the load scripts' rows (typically loaded from `.qvs` files)
  - One field containing a script identifier, such as the script file name, app name or app ID. Has to be unique across all scripts loaded into the app.

Download

1. Go to **Releases** and download the latest release ZIP (e.g. `qvsview-qs-v0.2.0.zip`).
2. Extract the downloaded ZIP. Inside you will find:
   - `readme.txt` — brief release notes
   - `LICENSE` — the MIT license
   - `README.pdf` — this documentation as a PDF
   - **`qvsview-qs.zip` — this is the actual extension file** that you upload to Qlik Sense

> ***Note:*** *The downloaded file is an outer ZIP that wraps the deployable extension ZIP. Extract the outer ZIP first, then use the inner `qvsview-qs.zip` for installation.*

Install in Qlik Sense

Client-managed (QSEoW):

1. Open the **Qlik Management Console (QMC)** → Extensions.
2. Click **Import**, select `qvsview-qs.zip` (the inner ZIP from the release package).
3. Open any app in the Sense hub, enter edit mode, and drag **QvsView.qs** from the custom objects panel onto a sheet.

Qlik Cloud:

1. Open the **Management Console** → Extensions.
2. Click **Add**, upload `qvsview-qs.zip` (the inner ZIP from the release package).
3. Open any app, enter edit mode, and drag **QvsView.qs** from the custom objects panel onto a sheet.

First Use

1. With the extension on a sheet in edit mode, open the **property panel** and add a **Dimension** — select the field that contains your script rows.
2. Add a second **Dimension** — select the field that contains the script identifier (e.g. file name, app name, or app ID).
3. Optionally adjust viewer settings: font size, line numbers, and other display options.
4. Switch to analysis mode. The script renders with full syntax highlighting.
5. Use the **section tabs** to jump between tabs, press **Ctrl/Cmd+F** to open the search bar, and click the fold indicators in the gutter to collapse sections.

Platform Support

Platform	Status
Qlik Sense Enterprise on Windows (client-managed)	Supported, 2025-May and later
Qlik Sense Cloud	Supported

Platform detection is automatic — the extension identifies the environment and adapts accordingly.

AI Analysis

QvsView.qs includes an optional AI-powered script analysis feature. Click the 🤖 **Analyze** toolbar button to send your script to a large language model for review. Results are rendered as Markdown with Mermaid diagrams and cached in `sessionStorage` for 30 minutes.

Supported Providers

Provider	Default endpoint	Auth	Best for
Ollama	`http://127.0.0.1:11434`	None required	Air-gapped / local
OpenAI	`https://api.openai.com/v1`	API key (Bearer)	Cloud
Anthropic	`https://api.anthropic.com/v1`	API key (header)	Cloud

Endpoints and LLM models can be customized in the property panel, allowing use of self-hosted models or API-compatible services.

Prompt Templates

Template	Focus
General	Summary, data flow, Mermaid flowchart, improvement ideas
Security	Credentials, injection risks, file access, data exposure
Performance	`WHERE` clauses, joins, resident loads, memory usage
Documentation	Comprehensive documentation with data model diagrams

Setup

1. Enable **AI Analysis** in the property panel under *AI Analysis → Enable AI Analysis*.
2. Choose a **provider** and configure the endpoint, model, and API key mode:
   - **Stored** — key saved in the Qlik object properties (convenient but visible to anyone with object access).
   - **Prompt at runtime** — key requested when you click Analyze and cached for the browser session.
3. Optionally set **Analysis scope** (current section or full script) and a **Custom system prompt**.
4. Click 🤖 **Analyze** in the viewer toolbar.

See docs/ai-analysis.md for full setup details, including Ollama CORS configuration for HTTPS environments.

Loading Experience

While waiting for the AI response, the modal shows:

- **Cycling humorous quotes** with smooth fade transitions (cycle time is configurable, 3–10 s)
- **Elapsed timer** tracking how long the analysis has been running
- **Snake mini-game** — a retro canvas-based game (WASD or arrow keys) to pass the time

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Architecture

Built with nebula.js and bundled as a single UMD file via Rollup (no dynamic imports).

Module	Description
`src/index.js`	Supernova entry point — data fetching, layout reactivity, event wiring
`src/syntax/highlighter.js`	Regex-based, stateful tokenizer with multi-line comment/string support
`src/syntax/keywords.js`	BNF-sourced keyword and function lists (400+ entries, 25 categories)
`src/syntax/tokens.js`	Token types and CSS color definitions (matching Qlik's native DLE colors)
`src/ui/viewer.js`	Full HTML renderer — section tabs, toolbar, line gutter, fold gutter, `<pre>`
`src/ui/search.js`	Search overlay: match finding, highlight rendering, folded-region auto-expand
`src/ui/ai-modal.js`	AI analysis modal: loading UX, Snake mini-game, Mermaid diagram rendering
`src/ai/`	Provider adapters, prompt templates, response caching, API key management
`src/ext/`	Property panel sections (viewer settings, toolbar, AI, about)

Contributing

Pull requests are welcome!

1. Fork the repository and create a feature branch.
2. Run `npm run lint:fix && npm run format` before committing.
3. Use Conventional Commits — `feat:`, `fix:`, `docs:`, `refactor:`, etc.
4. Open a pull request against `main`.

Report bugs or request features via GitHub Issues.

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